Class Activity 10.07.2026

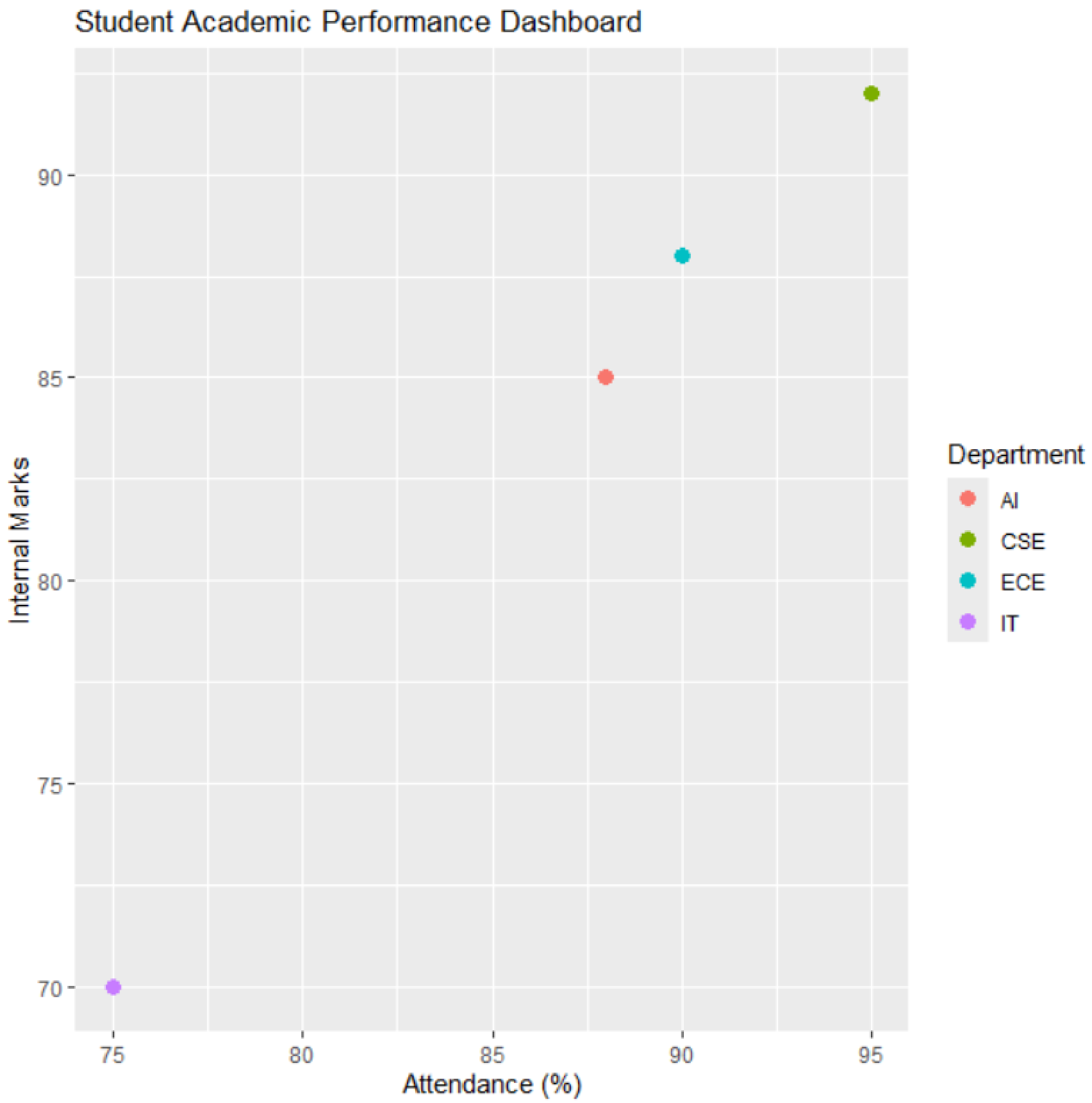
Scenario 1: Student Academic Performance Dashboard

```
library(ggplot2)

student <- data.frame(Attendance=c(95,88,75,90), Marks=c(92,85,70,88),
Department=c("CSE","AI","IT","ECE"))

ggplot(student, aes(x=Attendance, y=Marks, color=Department)) + geom_point(size=3) +

labs(title="Student Academic Performance Dashboard", x="Attendance (%)", y="Internal Marks",
color="Department")
```



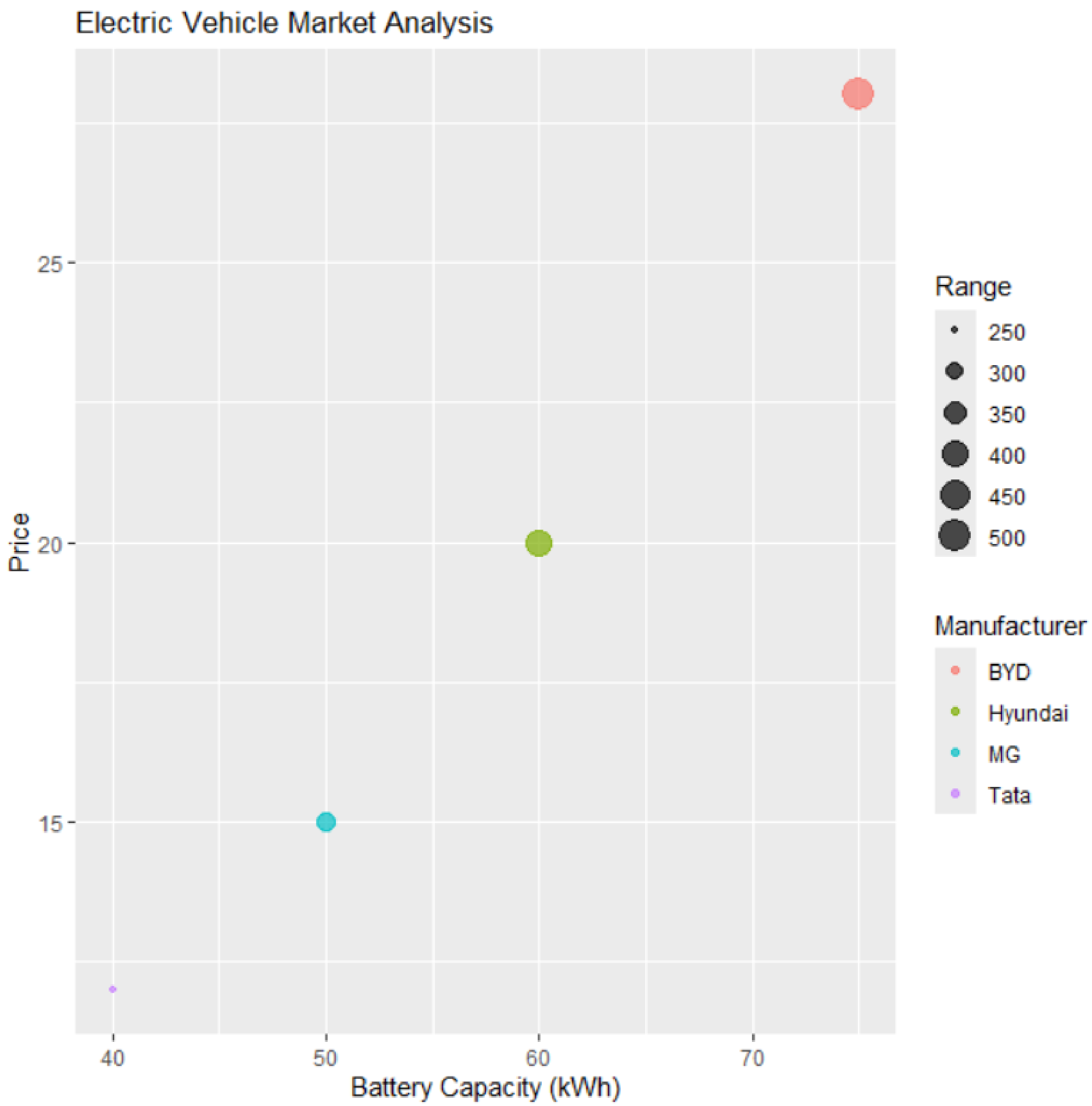
Scenario 2: Electric Vehicle Market Analysis

```
library(ggplot2)

ev <- data.frame(Battery=c(40,50,60,75), Price=c(12,15,20,28), Range=c(250,320,400,500),
Manufacturer=c("Tata","MG","Hyundai","BYD"))

ggplot(ev, aes(x=Battery, y=Price, size=Range, color=Manufacturer)) + geom_point(alpha=0.7) +

labs(title="Electric Vehicle Market Analysis", x="Battery Capacity (kWh)", y="Price")
```



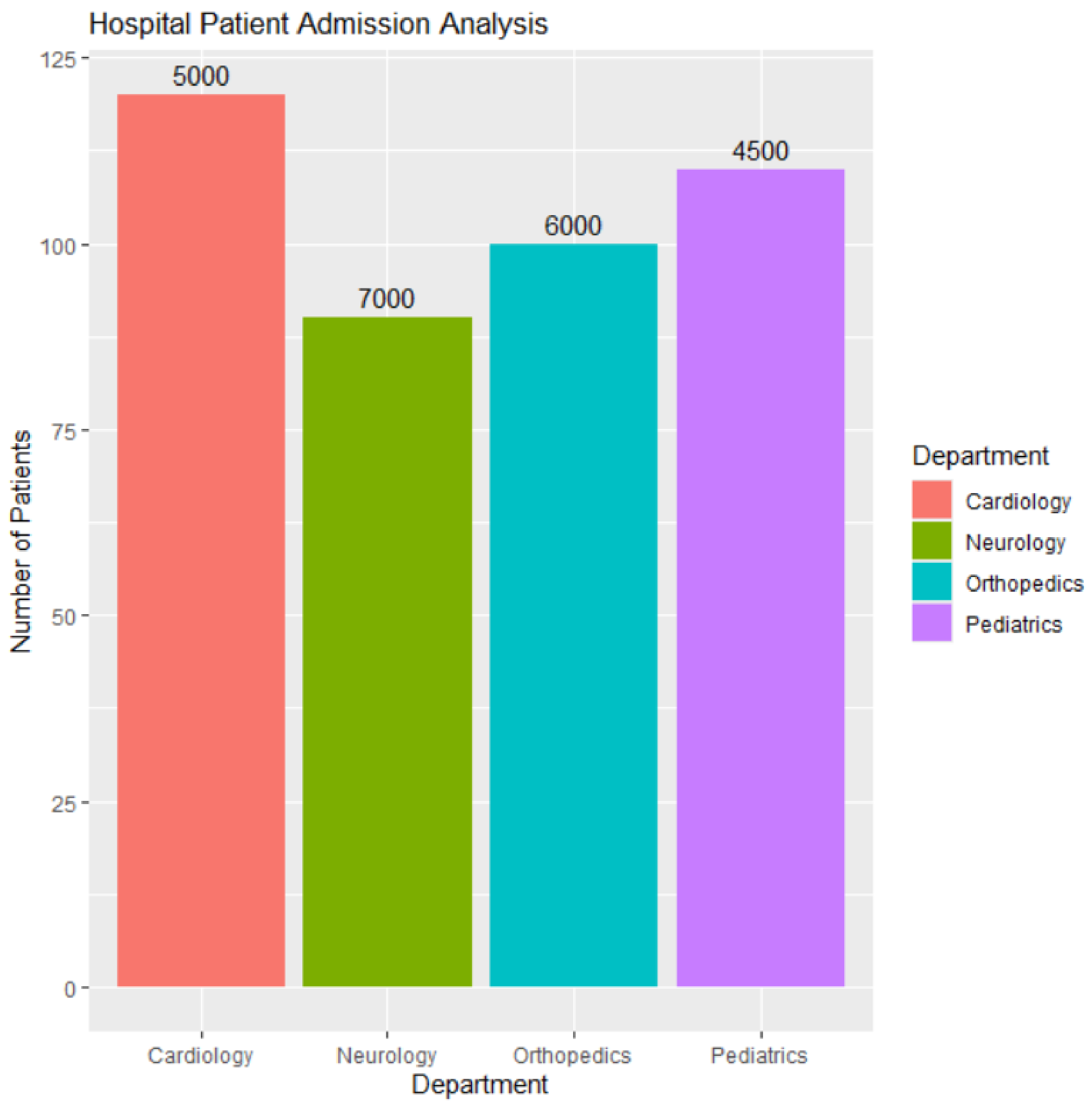
Scenario 3: Hospital Patient Admission Analysis

```
library(ggplot2)

hospital <- data.frame(Department=c("Cardiology","Neurology","Orthopedics","Pediatrics"),
Patients=c(120,90,100,110), Cost=c(5000,7000,6000,4500))

ggplot(hospital, aes(x=Department, y=Patients, fill=Department)) + geom_bar(stat="identity") +

geom_text(aes(label=Cost), vjust=-0.5) + labs(title="Hospital Patient Admission Analysis",
x="Department", y="Number of Patients")
```



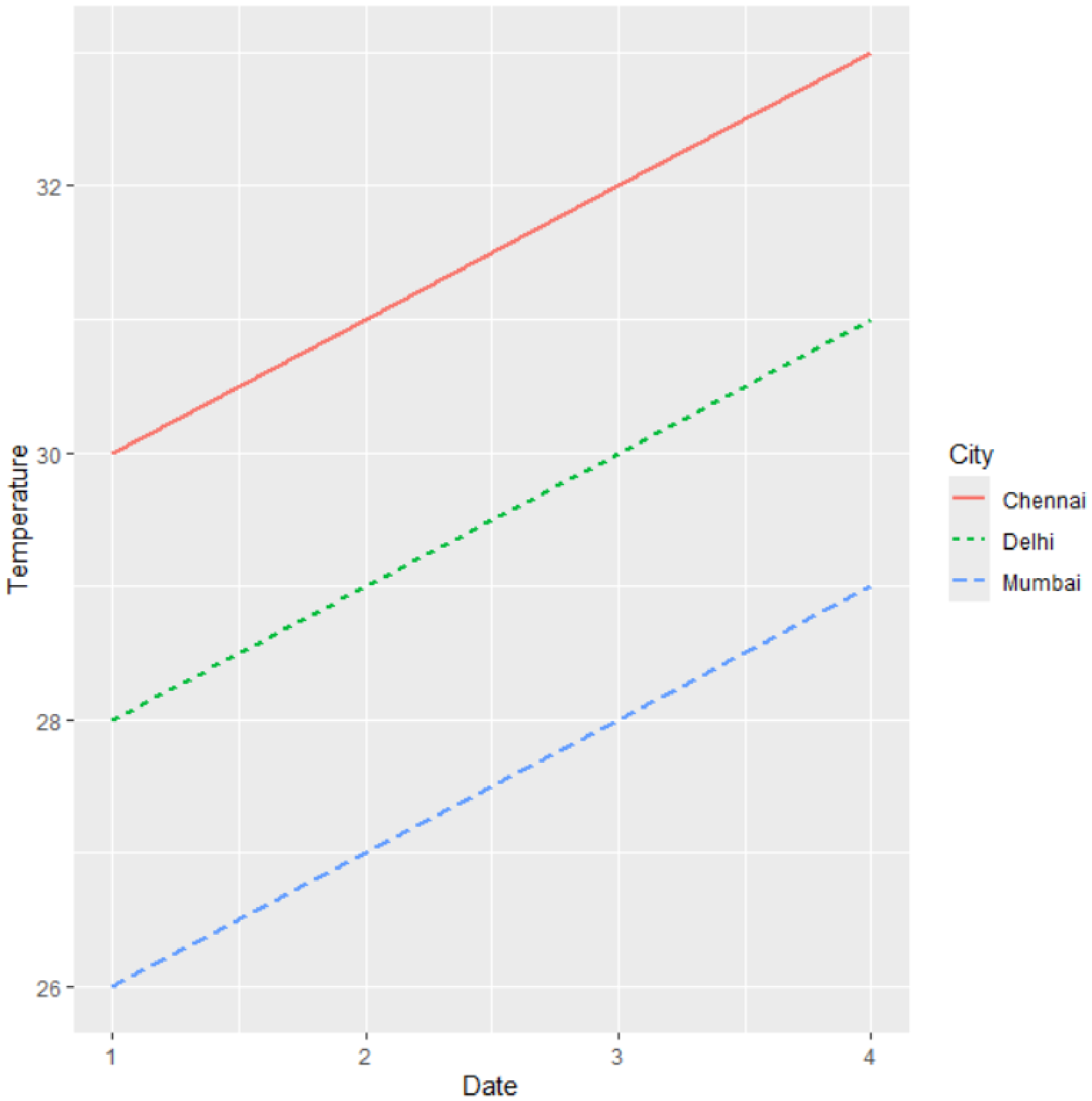
Scenario 4: Weather Monitoring System

library(ggplot2)

weather <- data.frame(Date=rep(1:4,3), Temperature=c(30,31,32,33,28,29,30,31,26,27,28,29), City=rep(c("Chennai","Delhi","Mumbai"), each=4))

ggplot(weather, aes(x=Date, y=Temperature, color=City, linetype=City, group=City)) + geom_line(size=1)

+ labs(title="Weather Monitoring System", x="Date", y="Temperature (°C)")



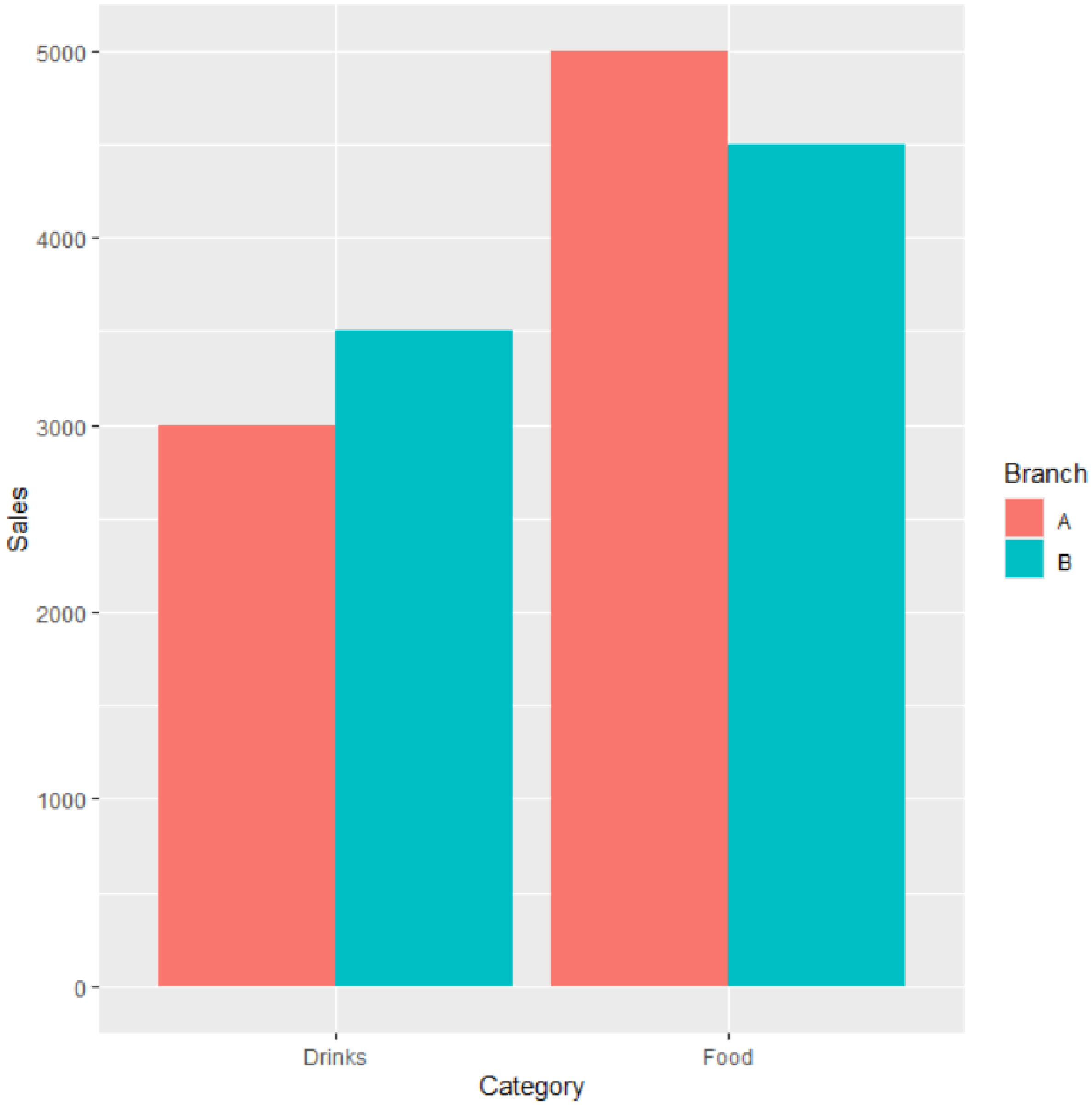
Scenario 5: Supermarket Sales Analysis

```
library(ggplot2)

sales <- data.frame(Category=c("Food","Food","Drinks","Drinks"), Sales=c(5000,4500,3000,3500), Branch=c("A","B","A","B"))

ggplot(sales, aes(x=Category, y=Sales, fill=Branch)) + geom_bar(stat="identity", position="dodge")

print(last_plot())
```



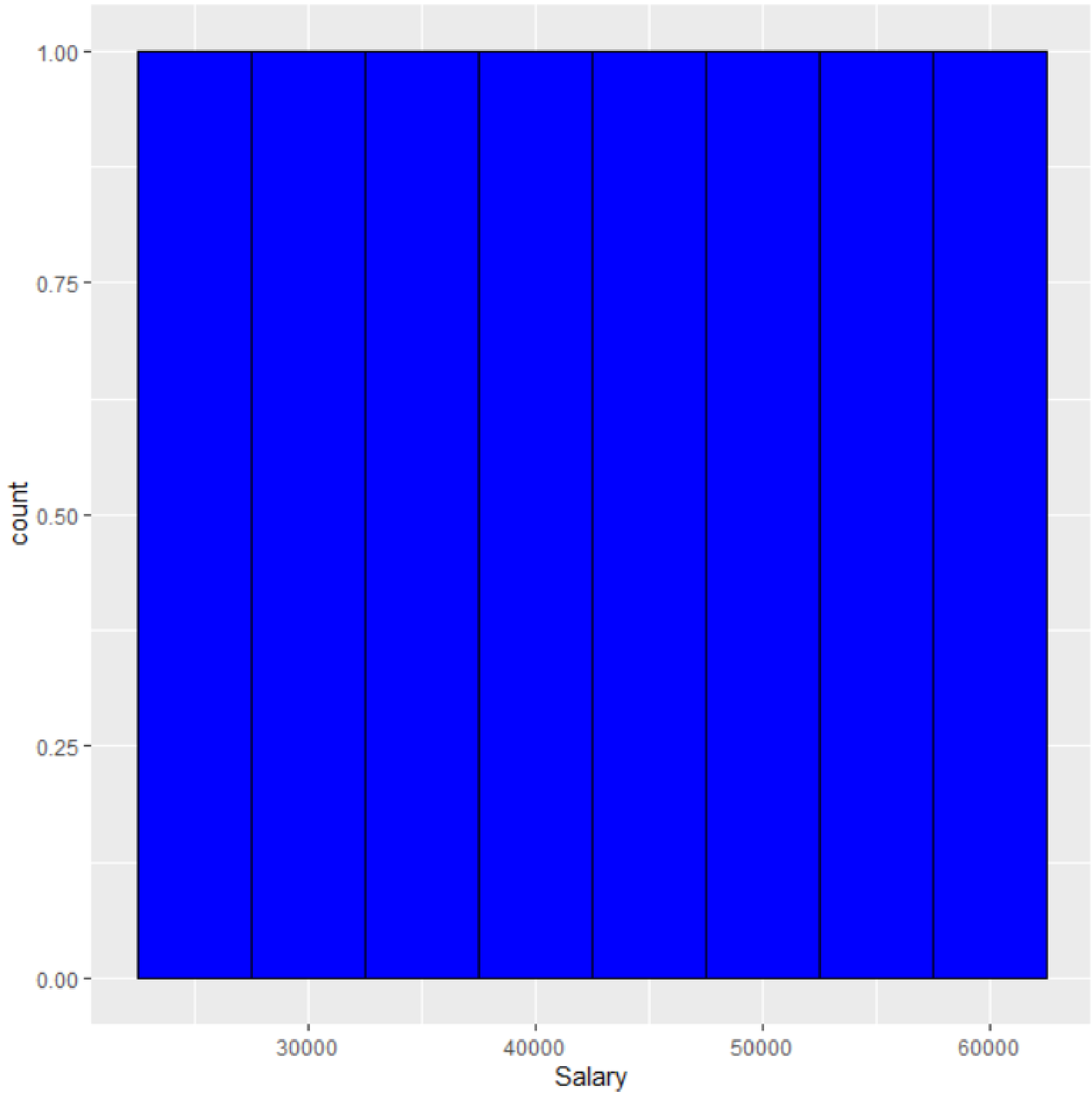
Scenario 6: Employee Salary Distribution

```
library(ggplot2)

employee <- data.frame(Salary=c(25000,30000,35000,40000,45000,50000,55000,60000))

ggplot(employee, aes(x=Salary)) + geom_histogram(binwidth=5000, fill="blue", color="black")

print(last_plot())
```



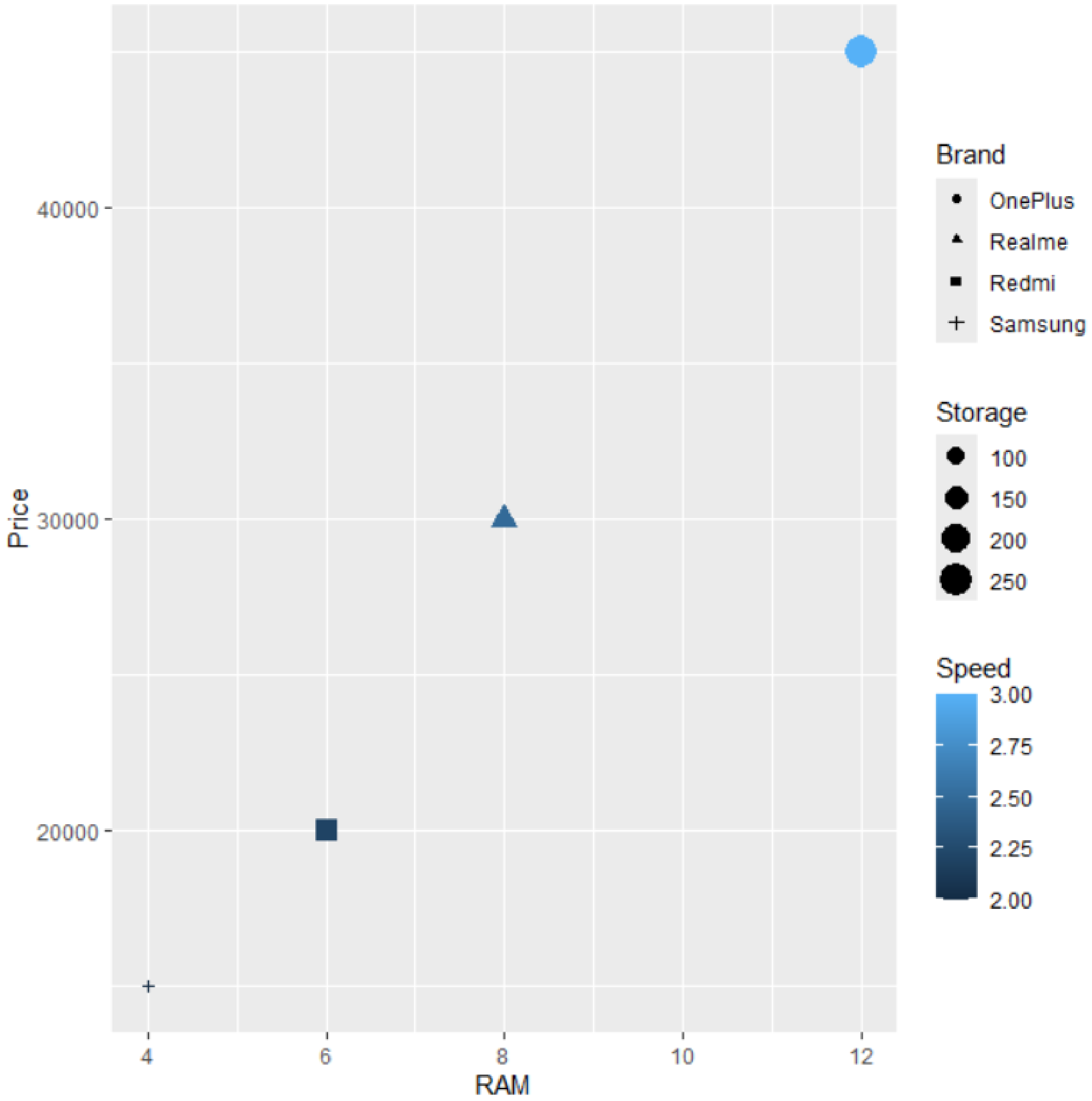
Scenario 7: Smartphone Market Comparison

```
library(ggplot2)

phone <- data.frame(RAM=c(4,6,8,12), Price=c(15000,20000,30000,45000),
Storage=c(64,128,128,256), Brand=c("Samsung","Redmi","Realme","OnePlus"),
Speed=c(2.0,2.2,2.5,3.0))

ggplot(phone, aes(x=RAM, y=Price, shape=Brand, size=Storage, color=Speed)) + geom_point()

print(last_plot())
```



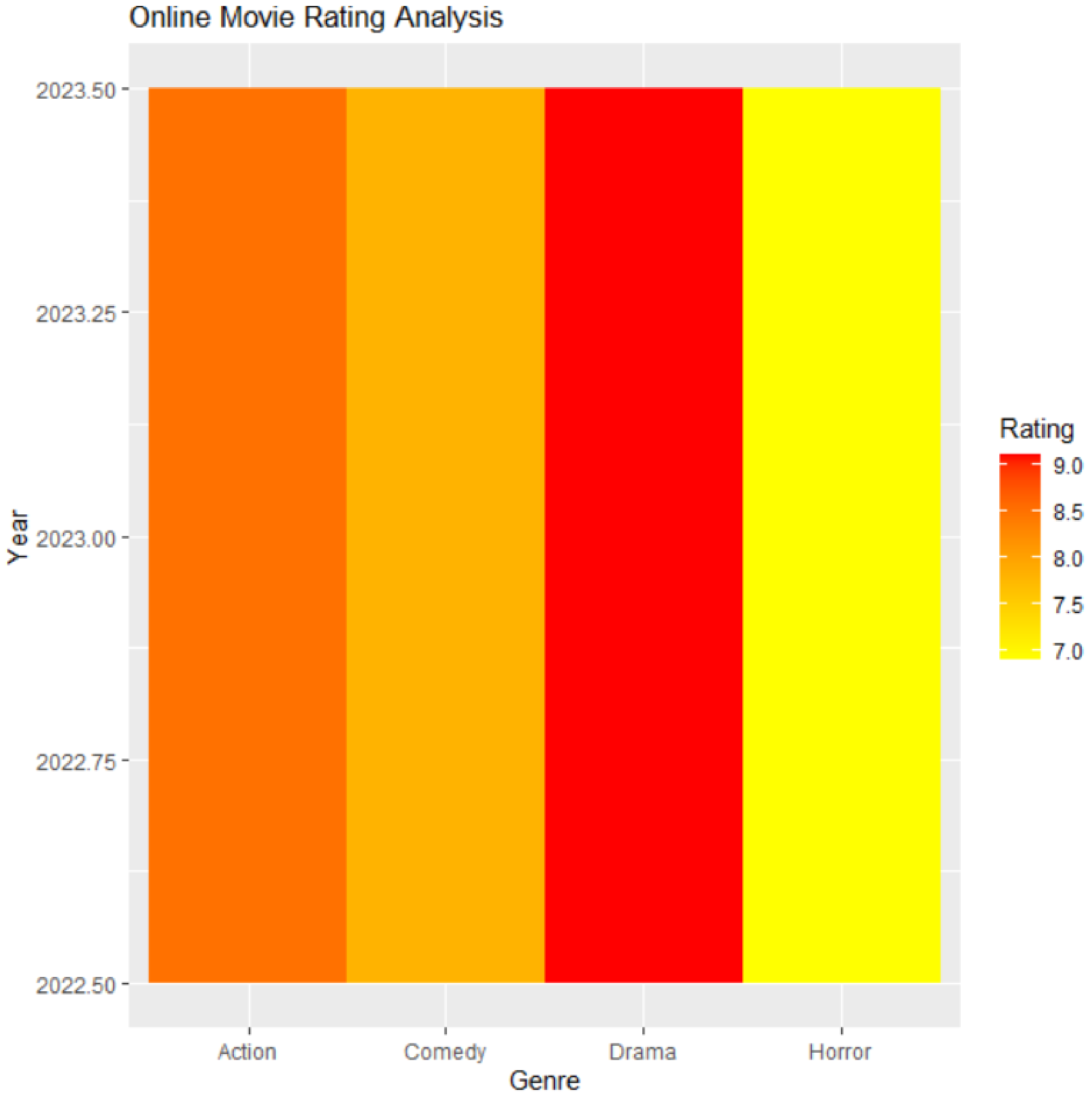
Scenario 8: Online Movie Rating Analysis

```
library(ggplot2)

movie <- data.frame(Genre=c("Action","Comedy","Drama","Horror"),
Year=c(2023,2023,2023,2023), Rating=c(8.5,7.8,9.1,6.9))

ggplot(movie, aes(x=Genre, y=Year, fill=Rating)) + geom_tile()

print(last_plot() + scale_fill_gradient(low="yellow", high="red") + labs(title="Online Movie Rating
Analysis"))
```



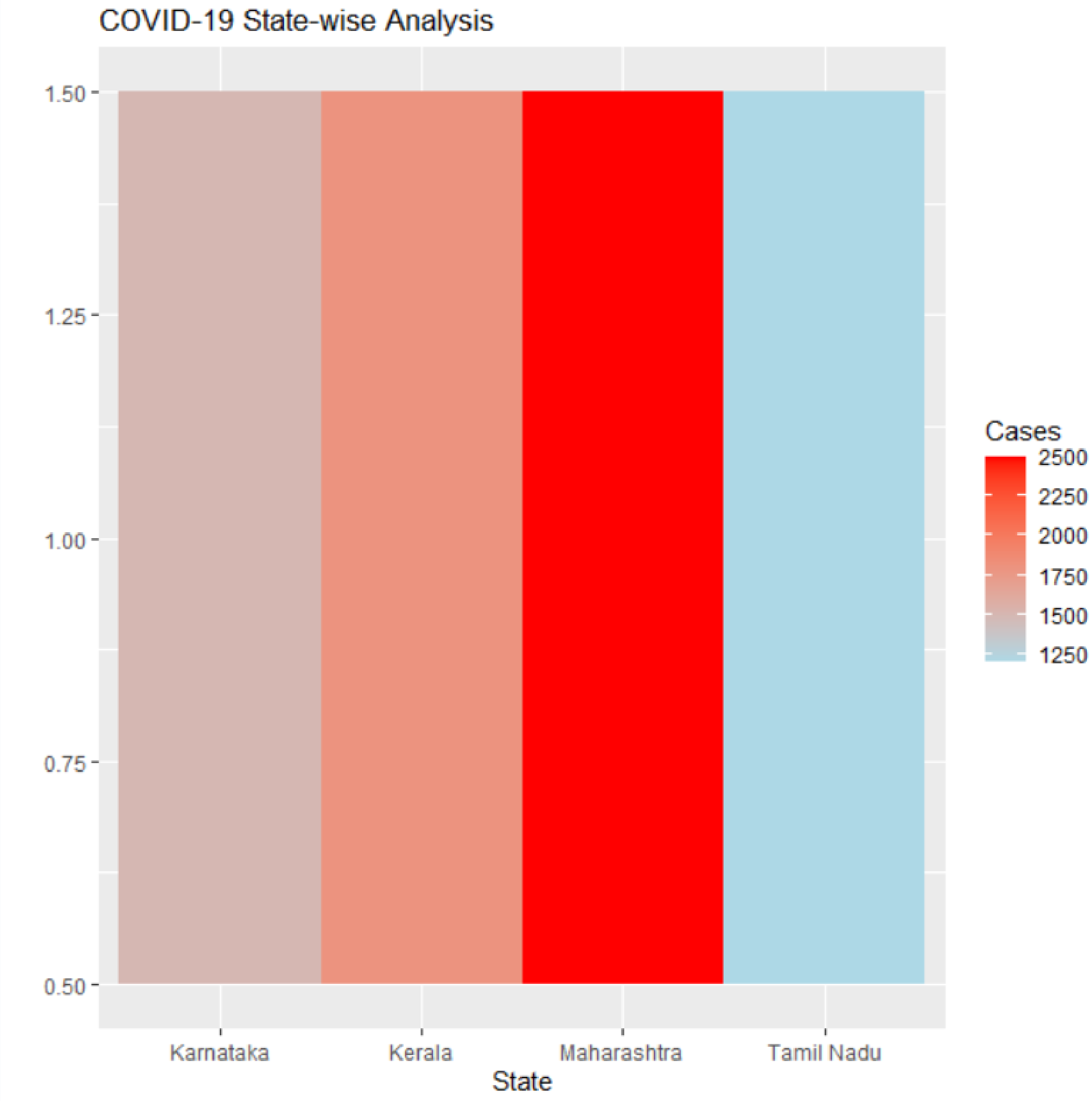
Scenario 9: COVID-19 State-wise Analysis

library(ggplot2)

covid <- data.frame(State=c("Tamil Nadu","Kerala","Karnataka","Maharashtra"), Cases=c(1200,1800,1500,2500))

ggplot(covid, aes(x=State, y=1, fill=Cases)) + geom_tile()

print(last_plot() + scale_fill_gradient(low="lightblue", high="red") + labs(title="COVID-19 State-wise Analysis", x="State", y=""))



Scenario 10: Iris Flower Classification

library(ggplot2)

ggplot(iris, aes(x=Sepal.Length, y=Petal.Length, color=Species, shape=Species)) +
geom_point(size=3)

labs(title="Iris Flower Classification", x="Sepal Length", y="Petal Length")

print(last_plot())